

Performance analysis of a stand-alone hybrid energy system for desalination unit in Egypt



Mostafa M. Ibrahim*, Nabil H. Mostafa, Ahmed H. Osman, Ahmed Hesham

Mechanical Power Engineering Department, Faculty of Engineering, Zagazig University, 44519 Zagazig, Egypt

ARTICLE INFO

Keywords:

Renewable energy
Hybrid power system
Desalination
Solar
Wind
Hydrokinetic

ABSTRACT

Renewable energy sources are economical systems suitable for seawater treatment systems wherever there is no electrical grid. This research focuses on finding the best performance of several combinations of these resources (off-grid). The performance compares two alternatives; the first consists of a photovoltaic solar panel, wind turbine, diesel generator, and battery. The second alternative consists of a photovoltaic solar panel, hydrokinetic turbine, diesel generator, and battery. All alternatives are used to drive a small desalination unit with a capacity of 1 m³/h. HOMER software is used to optimize the hybrid renewable energy system and perform techno-economic analyses. The optimum case from the first alternative combinations consists of a wind turbine with a rated power of 10 kW, diesel generator with a rated power of 4.90 kW, and 20 kW photovoltaic solar panel. This case achieves the minimum electrical power cost of 0.2252 \$/kWh and the corresponding minimum water cost of 1.10\$/m³. The optimum case from the second alternative combinations consists of a hydrokinetic turbine with a maximum power of 5 kW, diesel generator with a rated power of 4.90 kW, and 2.82 kW photovoltaic solar panel. This case achieves the minimum electrical power cost of 0.1216 \$/kWh and the corresponding minimum water cost of 0.56 \$/m³.

1. Introduction

Water and energy are the fundamental items in the present world for sustaining life. Water covers a great part of the globe in the form of seas, gulfs, and oceans. One of the real issues in the world nowadays is the developing drinkable water request [1]. The vast majority of the water on Earth is in the form of seawater or icebergs in the polar regions. Around 3% of the water on Earth is freshwater, whereas the remaining 97% is salty water. Uprooting the saltiness of seawater utilizing different desalination methodologies need the possibility to resolve the sweet water supply issue [2]. The two great methods of desalination are phase-based thermal processes and membrane processes based on the pressure difference. The essential thermal desalination methods utilized for freshwater contain a multi-stage flash and vapor compression strategies. The major popular membrane-based desalination techniques containing electrolysis and reverse osmosis (RO) [3]. Reverse osmosis is a pressure-driven separation procedure, where the saline result disperses its solvents by accidental dispersion over a film. Different desalination methods demand considerable amounts of energy to obtain potable water. As a result of industrialization, the feeding of energy has essential concentrate on centralized production depend on fossil fuels [4]. The utilization of renewable energy resources in

desalination methods reduces fossil fuel utilization and greenhouse gas emissions. This will help solve water shortages in separate and dry areas using nearby renewable resources [5].

Modeling a power generation system for a project is of fundamental importance for making choices and pre-feasible dissection. Many software tools are used to estimate which power systems can be obtained, such as HOMER [6]. Previous studies include few investigations into the integration of desalination with renewable energy strategies to produce freshwater.

Koreneos et al. [7] developed a model for desalination technologies coupled with renewable energy power systems to produce drinkable water at the lower potential expense. Their results showed that the calculation of the supplied energy will be smaller when using a renewable source, therefore, the calculation of the water preparation for the desalination unit will have a chance to be smaller. Setiawan et al. [8] investigated a design of a hybrid renewable energy power system and a diesel generator for RO desalination unit. They performed a simulation of a hybrid power system to find the optimum configuration with HOMER software. Their results demonstrated that the hybrid renewable energy power system submitted best performances for both power feeding and water supply.

Spyrou and Anagnostopoulos [9] developed a numerical algorithm

* Corresponding author.

E-mail address: dr.moustafa48@gmail.com (M.M. Ibrahim).

<https://doi.org/10.1016/j.enconman.2020.112941>

Received 17 February 2020; Received in revised form 3 May 2020; Accepted 4 May 2020

Available online 14 May 2020

0196-8904/ © 2020 Elsevier Ltd. All rights reserved.

Nomenclature

A	turbine area (m^2)
AC	alternating electric current (A)
COE	cost of energy (\$)
$C_{ann,tot}$	total annualized cost (\$/year)
C_{elect}	price of the electricity produced by the hybrid system (\$/kWh)
$C_{p,H}$	coefficient of the hydrokinetic turbine performance
d	day number of the year
DC	direct electric current (A)
DG	diesel generator
f	expected inflation rate
F_1	slope of the fuel curve.
F_0	intercept coefficient of the fuel curve
F_{PV}	derating factor
G_T	solar radiation incident on the PV array (W/m^2)
$G_{T,STC}$	standard irradiation (W/m^2)
i	real discount rate
i^l	nominal discount rate
n	project lifetime (year)
P_G	electrical output of the generator (kW)
P_{HKT}	power generated from the hydrokinetic turbine (kW)
P_p	pump pressure (bar)
$P_{PV-rated}$	PV array rated capacity (kW)
P_R	rated capacity of the generator (kW)
P_{RO}	total power need for RO unit (W)
P_{WT}	wind turbine power output (kW)
$P_{WT,STP}$	wind turbine power output at standard conditions (kW)
T_{amb}	ambient air temperature ($^{\circ}\text{C}$)
T_C	PV cell temperature ($^{\circ}\text{C}$)
$T_{C,STC}$	PV cell temperature under standard conditions ($^{\circ}\text{C}$)

U_{anem}	wind speed at anemometer height (m/s)
U_{hub}	wind speed at the hub height of the wind turbine (m/s)
V	water current velocity (m/s)
Z_{anem}	anemometer height (m)
Z_{hub}	hub height of the wind turbine (m)
α	elevation angle
α_p	temperature coefficient
β	tilt angle
δ	declination angle, [degree].
η_{HKT}	combined efficiency of the hydrokinetic turbine and the generator
η_{HP}	motor and pump efficiency
η_{inv}	inverter efficiency
η_{wire}	wire efficiency
P	air density (kg/m^3)
ρ_0	air density under standard conditions (kg/m^3)
ρ_w	water density (kg/m^3)

Abbreviations

CRF	capital recovery factor
HKT	Hydrokinetic Turbine
HOMER	Hybrid Optimization Model for Electric Renewable
HRES	Hybrid Renewable Energy System
NOCT	Normal Operating Cell Temperature ($^{\circ}\text{C}$)
NPC	Net Present Cost (\$)
PV	Photovoltaic
RE	Renewable Energy
RO	Reverse Osmosis
TDS	Total Dissolved Solids
WHO	World Health Organization.
WT	Wind Turbine

to examine the performance of a hybrid PV solar and wind energy power system for RO desalination units to supply drinkable water for the Greek islands. Their hybrid desalination system was able to achieve the water request with a particular water preparation expense (1.5–3.0 €/m³). Cherif and Belhadj [10] proposed a hybrid renewable (wind and PV solar) power system for RO desalination unit in southern Tunisia. Their effects demonstrated that utilizing a hybrid renewable power system for RO desalination unit to prepare drinkable water from salty water was a suitability method for Tunisia. Khalifa [11] investigated a hybrid energy system (PV solar, wind, and diesel generator) to supply the required electricity for the RO desalination unit in Iraq. Their investigation demonstrated that the lowest expense of water 1.8 \$/m³ for desalination unit powered by either a diesel generator or by a combination of PV solar, wind, and diesel generator. When the desalination unit relies on renewable energy only, water costs have risen to 2.3–2.4 \$/m³. Hossam-Eldin et al. [12] investigated the utilization of hybrid energy systems coupled with RO desalination unit. They developed a computer program to choose the better hybrid renewable energy systems integration that generating potable water in a comparatively economic expense. Their study explained that hybrid systems considered as costly power resources when utilized in power implementations for remote sites. Mentis et al. [13] developed a tool able to design and size optimally both desalination and renewable energy units for several Greek islands. Various parameters related to the desalination's energy prerequisites are taken into accounts such as input information which interests mechanical transformation performance, resource accessibility and economic information. Their results indicated that the water selling price ranges from 1.45 €/m³ to 6 €/m³. Javed et al. [14] proposed a hybrid energy system composed of a wind turbine, PV solar, and battery to calculate the optimal number of the wind turbine, PV solar modules, and batteries for off-grid island. Their results

had good acceptance when compared with HOMER software for four different cases. Their model studied the execution of the hybrid system and analyzed the impacts of renewable energy resources and a variety of loads on the system economic. Krishan and Suhag [15] introduced a techno-economic analysis and optimum planning of hybrid renewable energy systems to meet up with the domestic and agricultural electric load necessities. Their study introduced three separate configurations such as; wind/battery, PV/battery and wind/PV/battery using HOMER software. Their results demonstrated that wind/PV/battery was the majority expense-effective arrangement for that particular area.

Hydrokinetic power generation can be considered a modern kind of hydropower that generates power from the kinetic energy of streaming water. resolved the utilization of independent hydrokinetic systems for country electrification. As these stations need to place the turbines in the path of the moving water in the rivers which depend on the stability of the flow to operate the water turbines. Therefore, there is a lack of available literature looking at how to use this type of technology with traditional sources such as diesel generators or renewable sources such as wind and solar energy.

Kusakana [16] proposed the utilization of hydrokinetic power integrated with hybrid systems for continuous electrical power feeding isolated load in South Africa. They used HOMER software to modeled and simulated different hybrid system configurations. Their results from two different investigations showed that the hybrid system that contained a hydrokinetic module had a lower cost when compared with the other system not contained a hydrokinetic module. Kusakan and Vermaak [17] investigated the utilization of a hybrid system composed of diesel generator and hydrokinetic turbine for continuous electricity production in South Africa. The results showed that the hybrid system (diesel and hydrokinetic turbine) had less expense of energy contrasted with different supply alternatives, for example, standalone PV,

hydrokinetic or diesel generator. Lata-García et al. [18] presented a hybrid power system, which consists of PV, HKT, hydrogen, and batteries to supply electric power to a group of isolated loads. The hydrokinetic turbine is optimally positioned in a waterway cross-section for optimum hydroelectric generator behavior. Their results demonstrated that the hydrogen subsystem has a maximum cost of 70,234\$ (50.64%) and produces power by 3.96% of the total power. Azmi and Ramli [19] introduced a techno-economic and ecological dissection of various hybrid systems for feeding electricity to Iraqi country village using HOMER code. Their results showed that the hybrid PV/hydro/diesel/battery was the best economical choice with a net present expense of 113,201 \$ and it was appropriate at technical and environmental performance levels.

Dependent upon the previous studies, there have been few studies on hybrid renewable energy systems that are used in desalination, especially systems that include a hydrokinetic turbine. These systems considered both a prediction model and techno-economic model to enable the integration of renewable energy sources into the desalination-fresh water production systems. The integration between energy and desalination should be further investigated to achieve minimum pollution and low cost for the freshwater production process, especially models that contain a hydrokinetic turbine.

The main aim of this study is to assess the cost-effectiveness two different options of hybrid power systems (off-grid) using HOMER software for a small RO desalination unit in Ras El Bar, Egypt north. The levelised cost method is applied to examine the unit energy and water costs in terms of various economic parameters for two different options. The first option components are; diesel generator (DG), wind turbine (WT), photovoltaic panels (PV), and battery. The second option components are; diesel generator (DG), hydrokinetic turbine (HKT), photovoltaic panels (PV), and battery. These components replaced the wind turbine (WT) component by the hydrokinetic turbine (HKT). Each one of the two options has seven configurations of the hybrid system. In each option, the following methodology has been used:

- Choosing appropriate energy system components.
- Optimizing the hybrid power system using HOMER software.
- Economical study for each case of the two options with respect to the cost of energy produced (COE) and the cost of desalinated water (LCOW).
- Calculate emissions from each case studied and the impact on the environment.
- Study the effect of the diesel fuel price on the desalinated water production cost.

The work on this paper is organized as follows; Section 1 presents a literature review for the previous work of hybrid systems. In the second section, the methodology approach to investigate the hybrid energy system components for RO unit with necessary mathematical model. Section 3 introduces the economic model for each case of the two options with respect to the cost of energy produced (COE) and the cost of desalinated water (LCOW). Section 4 introduces the results and discussion for the two proposed options. Finally, Section 5 contains the conclusions for the current study.

2. Methodology approach and hybrid energy system components

The proposed desalination system is concerned with the integration of the power generation system with the RO desalination unit. The RO desalination unit contains a capacity tank, membrane, and pump. This combination plans to obtain about 1 m³/h to generate potable water using different units to generate the necessary energy. Every person needs about 20 L from freshwater a day as recommended by UN organizations and the World Health Organization (WHO) [20]. The water from the desalination unit discussed in the current work might help the everyday request to 1200 person. The specifications of the reverse

Table 1

Technical specification of RO desalination unit.

Characteristic	Unit	Value
Feed Pressure	Bar	40.23
Pump efficiency	–	0.85
Feed Flow	m ³ /h	3.333
Fouling factor	–	0.85
Permeate Flow	m ³ /h	1
Recovery	%	30
Power demand	kW	4.38
Energy consumption	kW.h/m ³	4.38
Membrane	–	SW30HRLE-370/34i
Membrane numbers	–	6
Permeate TDS	mg/L	433.93

osmosis desalination unit contained in the present work are registered in Table 1. Table 1 statements that the water goodness and specific energy-consuming of operation depend upon the feed water goodness, membrane kind, and container structure. Conducting this study is based on modeling the optimal energy system using HOMER program [21] and calculating the cost of water production from desalination. HOMER is an optimization tool for the design of power systems, which makes it possible to allow it to improve the size of its components by conducting an economic technical examination. In this software, an extensive variety of applications that incorporate PV panels, wind and small-hydro turbines, fuel cells, batteries, generators, and boilers could be demonstrated toward recognizing considering stand-alone and grid-connected systems over their lifetime. In this program, a wide range of power generation systems that incorporate wind turbines, small hydroelectric, PV panels, batteries, fuel cells, and diesel generators can be designed. Fig. 1 shows the detailed steps of the current study analysis.

2.1. Location characterization

The current RO studied system was proposed to be installed on Ras El Bar city (31.5°N, 31.8°E) in the north of Egypt. Ras El Bar is located on a peninsula on the shore of the Mediterranean Sea and is bounded by the western part of the Damietta River branch. The area of “Lessan” is located in the far northern part of the peninsula. At this point the arm of the Damietta river flows into the Mediterranean Sea, giving Ras El Bar the shape of a triangle [22]. Fig. 2 shows the average monthly wind speed for Ras El Bar city. The maximum wind speed is 6.27 m/s in February while the average wind speed during the year is 5.35 m/s. Fig. 3 shows the average monthly temperature which indicates that August is the hottest month with an average temperature of 26.94 °C. The coldest month is January at 15 °C with the most daily sunshine hours at 14 in July. The clearness index is a factor utilized to predict the influence of clouds on extraterrestrial radiation which depends on time. The clarity index is used to predict the amount of solar radiation falling on a horizontal surface at any location on the Earth's surface. Fig. 4 shows both the average monthly solar radiation information and the values of the solar radiation's clearness index for the same format according to the NASA (National Aeronautics and Space Administration) database.

The maximum solar irradiance is in June (8.5 kWh/m²/day) while the minimum solar irradiance is in December (2.97 kWh/m²/day) with annual average solar radiation at 5.88 kWh/m²/day. The average water speed through the Damietta Nile branch for Ras EL Bar city is almost constant throughout the year about 1.4 m/s [23]. The compositions of the seawater for Ras El Bar City, Egypt, and the principles for potable water stated by WHO are recorded in Table 2 [24].

2.2. Seawater reverse osmosis system (SWRO)

The reverse osmosis subsystem incorporates a pump, membranes, and a capacity tank. Fig. 5 demonstrates a schematic outline of a small-

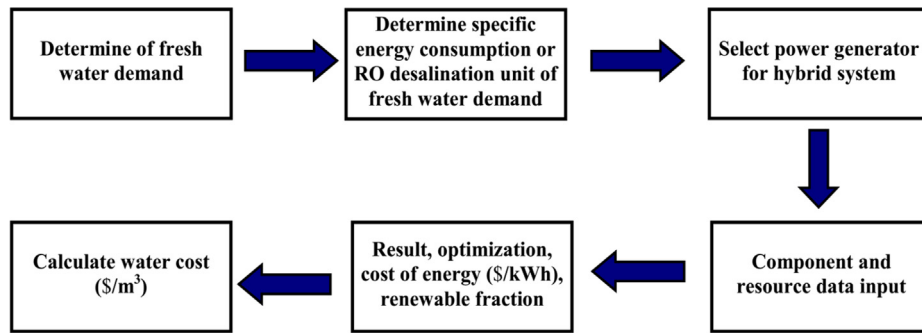


Fig. 1. Detailed steps of the analysis in the current study.

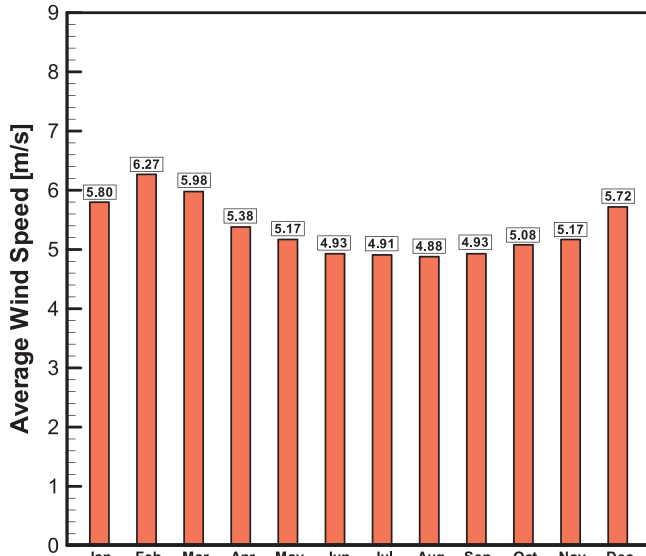


Fig. 2. The average monthly wind speed data at Ras El Bar city.

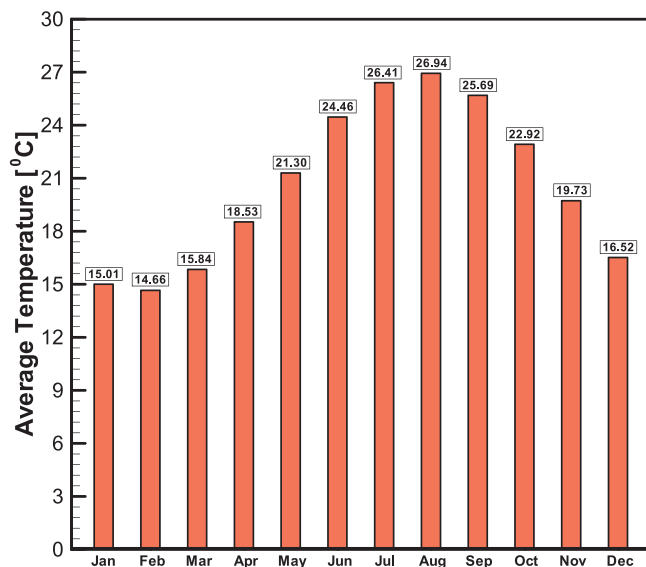


Fig. 3. The average monthly temperature data at Ras El Bar city.

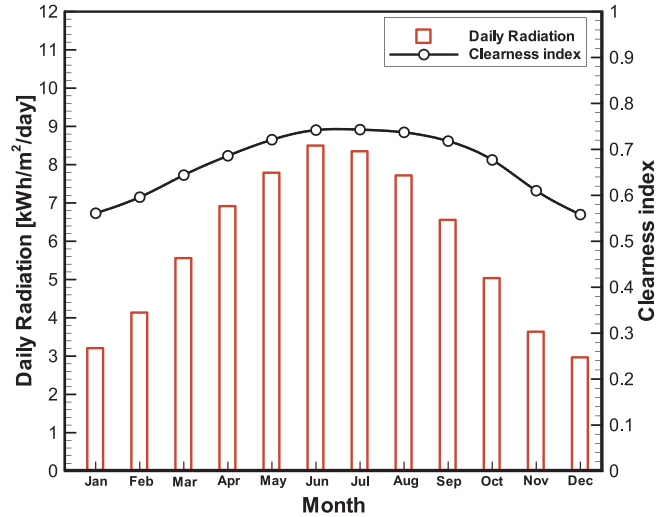


Fig. 4. Monthly average daily solar radiation and clearness index at Ras El Bar city.

Table 2

Seawater compositions and potable water standard [24].

Characteristic	Unit	Value for studied site	WHO standard
Conductivity	μS/cm	64,340	–
PH	–	8.2	> 6.5 and < 9.5
COD	mg/L	< 50	–
K ⁺	mg/L	600	12
Na ⁺	mg/L	11,515	200
Mg ²⁺	mg/L	1430	50
Ca ²⁺	mg/L	440	< 100
CO ₃ ²⁻	mg CaCO ₃ /L	20	–
HCO ₃ ⁻	mg CaCO ₃ /L	135	–
Cl ⁻	mg/L	21,520	250
SO ₄ ²⁻	mg/L	2205	500
TDS	mg/L	37,864.4	< 1000

selected as FILMTEC SW30HRLE-370 / 34i, which works with the best performance for its lifetime. The membrane component chosen for the desalination plant has many properties, including a high removal element, high throughput of potable water for utilization in harsh feeding water conditions, smooth birch processes, including low water costs. For the desalination process utilizing the reverse osmosis method, the energy sources consumed in the desalination plant are one of the economic fundamentals of the operation of the plant. The energy requirements of the RO desalination unit involve both pumping energy and chemical treatment. The total energy required for the pumping process is calculated by [20]:

scale RO unit that expects 1 m³/h of drinkable water preparation. The reverse osmosis system consists of membranes, pump, and a capacity reservoir. Fig. 5 shows a sketch of the basic components of the RO desalination unit expecting 1 m³/h of drinkable water preparation. The membrane that performs the separation of salts from water has been

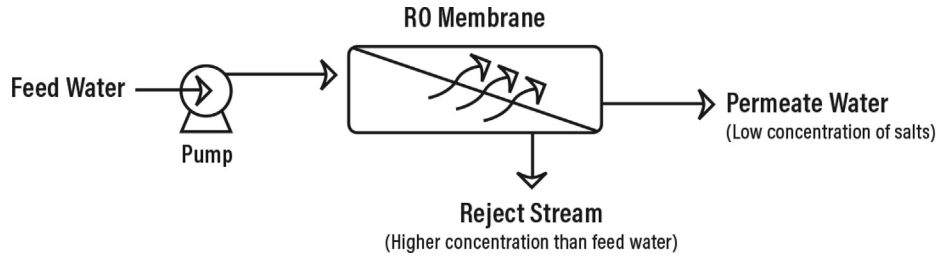


Fig. 5. Reverse osmosis system.

$$P_{RO} = \frac{27.77 Q_f P_p}{\eta_{HP}} \quad (1)$$

where: P_{RO} = Total power need for the RO unit, [W]

2.3. Power from PV panels

Photovoltaic solar cells are used as part of a hybrid grid-off system to provide energy to the RO desalination plant in the present work. Energy generates from PV solar cells in HOMER has been computed utilizing Eq. (2) [21,25].

$$P_{PV} = P_{PV-rated} \cdot F_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_p (T_C - T_{C,STC})] \times \eta_{inv} \times \eta_{wire} \quad (2)$$

where:

$P_{PV-rated}$ = Average energy from PV solar cells, [kW]

$G_{T,STC}$ = The solar radiation at standard conditions, [1000 W/m²]

α_p = The coefficient of temperature [% / °C]

The temperature of the PV cell can be computed utilizing the energy balance for the PV array:

$$T_C = T_{amb} + \left[\frac{NOCT - 20}{800} \right] \cdot G_T \quad (3)$$

where: T_{amb} = The ambient air temperature, [°C]

The capacity of PV solar cells is determined to utilize HOMER's automatic size technique. The power output of the PV solar modules depends on its tilt angle; Therefore its tilt angle must be selected to obtain maximum benefits from the PV panels. The declination angle (δ) must be determined from the relation [26]:

$$\delta = 23.45 \cdot \sin \left[\frac{360}{365} (284 + d) \right] \quad (4)$$

where: d = The day number of the year

The elevation angle (α) can be calculated from the relation [26]:

$$\alpha = 90 - \phi + \delta \quad (5)$$

where: ϕ = The latitude (for Ras El Bar; $\phi = 31.5$)

The tilt angle (β) of the panel is calculated from the relation [26]:

$$\beta = 90 - \alpha \quad (6)$$

The PV panels are fixed by 45° inclination angle without a tracking system.

2.4. Wind turbine system

The electrical energy generated from a wind turbine will be computed with the following steps. First, the wind speed at the center of the turbine hub height can be computed by utilizing the logarithmic theory. Next, the effect of standard air density is computed at the center of the turbine hub height and its impact on the energy produced by the wind turbine. Finally, the actual air density is computed and its effect on energy production from wind turbine [27]. The wind speed at the

turbine hub height is estimated by HOMER using the following relationship [21]:

$$\frac{U_{hub}}{U_{anem}} = \frac{\ln(Z_{hub}/Z_0)}{\ln(Z_{anem}/Z_0)} \quad (7)$$

where U_{hub} is the wind velocity at the turbine hub height, U_{anem} is the wind velocity at anemometer height, Z_{hub} is the turbine hub height, Z_{anem} is the anemometer height.

Wind turbine performance under standard temperature and pressure conditions (STP) is determined from the wind turbine energy characteristic curve. To adapt the power value of the wind turbine predicted by the energy curve to the actual air density, the HOMER program multiplies the power according to the following relationship:

$$P_{WT} = \left(\frac{\rho}{\rho_0} \right) P_{WT,STP} \quad (8)$$

where ρ is the actual air density, ρ_0 is the air density at standard conditions, P_{WT} is the output wind turbine power, and $P_{WT,STP}$ is the wind turbine power at standard conditions. In the current work, the capacity of the selected wind turbine was 10 kW with the power curve illustrated in Fig. 6.

2.5. Hydrokinetic turbine (HKT)

Hydrokinetic Turbine (HKT) is a turbine that extracts power from the water kinetic energy in open channels. It is considered a clean source to supply energy, based on the utilize of running water in oceans and rivers [28]. This type of energy source is utilized to convert energy with low effect environmental. Various researchers have proposed several types of hybrid systems depend on renewable sources available

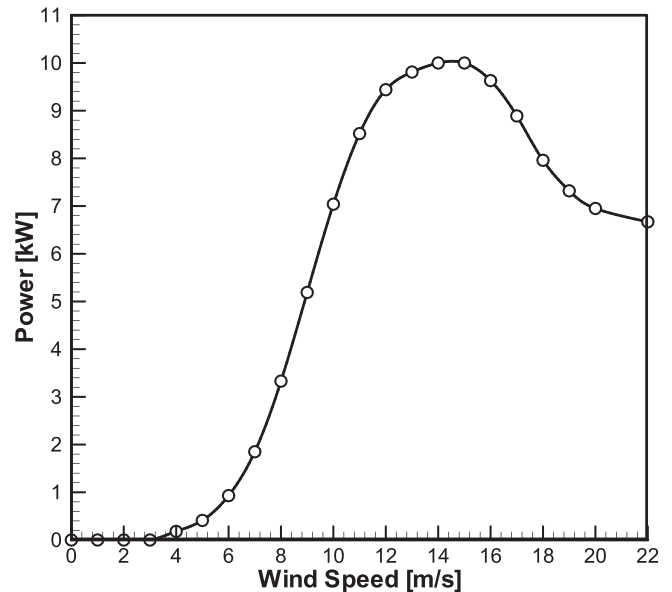


Fig. 6. Power curve for wind turbine.

especially in rivers that flow into the sea where they have a two-way flow of water. However, renewable energy research till now has focused on systems modeling, simulation, and optimization [29].

The power produced by the hydrokinetic turbine (P_{HKT}) is expressed as [30]:

$$P_{HKT} = \frac{1}{2} \rho_w \cdot A \cdot V^3 \cdot C_{P,H} \cdot \eta_{HKT} \quad (9)$$

where ρ_w is the water density, $C_{P,H}$ is the hydrokinetic turbine performance coefficient, η_{HKT} is the hydrokinetic system efficiency, V is the water speed, and t is the time.

The HKT model chose in the present study is the Smart Free Stream Turbine. This turbine is installed on the bed of the river or canal. The turbine rotor is a curved blade which is used to improve performance against debris. The permanent-magnet underwater generator provides AC power. The maximum output power from the HKT turbine is 5 kW, and the underwater permanent-magnet generator provides three-phase AC power. The characteristics of the HKT depend on the speed of the river as shown in Fig. 7.

2.6. Diesel generator model

As a result of renewable energy sources of a non-continuous nature, diesel generators should be used in hybrid power systems to ensure continued energy production. The next equation gives the rate of using the fuel consumed in the diesel generator in terms of the output power generated [21]:

$$FC_G = F_0 \cdot P_G + F_1 \cdot P_R \quad (10)$$

where P_G is the power production of the diesel generator, P_R is the diesel generator rated capacity, F_0 is the fuel intercept coefficient, and F_1 is the fuel curve slope. The two coefficients have the values of $F_1 = 0.08145 \text{ L/kWh}$ and $F_0 = 0.2460/\text{kWh}$, That was used in the current study [25]. The power of the diesel generator is estimated using HOMER's automatic size technique.

2.7. Battery energy storage system

Batteries would be utilized for energy backup due to the discontinuous nature of energy production from renewable energy resources. Batteries are utilized to benefit from excess energy from renewable energy sources (PV cells or wind turbines) used to charge batteries. Also, batteries are used when the required load is less than the energy generated from the available sources where the energy from the charged batteries is supplemented [31,32]. In the current study, a battery with 1 kWh of energy storage was chosen from the type of a generic 12-volt lead-acid battery.

2.8. Converter

A converter is an equipment that deals with electric current where it does the inversion process, which converts DC to AC. Also, Converter performs the rectification process which converts AC into DC. The converter capacity is estimated using HOMER's automatic size technique.

3. Economic model and hybrid power systems

3.1. Economic model

Economic analysis is an important stage to achieve the optimal energy production configuration from the proposed hybrid systems. The economic estimation of the power plants in HOMER software depends on the calculation of the net present cost (NPC) of the plant components. The NPC of a system is the variance between the present value of both all revenue earned from the system and the present value of all

expenses throughout the lifetime of the system. The NPC computed utilizing Eq. (11) [21,33].

$$C_{npc,tot} = \frac{C_{ann,tot}}{CRF} \quad (11)$$

where $C_{ann,tot}$ is the overall cost(\$/year), and CRF is the factor of capital recovery. The CRF can be computed utilizing Eq. (12) [34].

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (12)$$

where n is the lifetime of the project, and i is the rate of real annual interest (%). The rate of real interest is computed from the nominal interest rate using the following [21,33]:

$$i = \frac{i' - f}{1 + f} \quad (13)$$

where f is the rate of inflation, and i' is the rate of nominal interest. Both discount rate and inflation are determined based on the life of the project, in this study, the project lifetime considered 25 years with an annual decrease of 1% and the discount rate was set at 8% and inflation rate at 2%. HOMER software computes the levelized of energy cost (LCOE) as the cost calculation per kWh of electrical production energy from the energy system. To compute levelized of energy cost, the total cost of electricity produced annually is divided by the amount of electricity produced during this period. The LCOE computed using Eq. (14) [21,33].

$$LCOE = \frac{C_{ann,tot}}{E_{ser}} \quad (14)$$

where $C_{ann,tot}$ is the totality of the annualized expenses of each part in the hybrid power system (\$/year), and E_{ser} is the electricity served (kWh/yr). The economic and technical specifications of the hybrid power system utilized in the modeling are described in Table 3 [35,36].

3.2. Hybrid power generation system options

In this study, the power supply available options of several components may be used with a single component alone or a combination of more than one component to form hybrid systems for power generation. The battery is integrated into the power generation system where it stores excess power and feeds load when there is a shortage of power supply [37]. The main components available that can be integrated into various hybrid configurations are DG, WT, PV, HKT, and

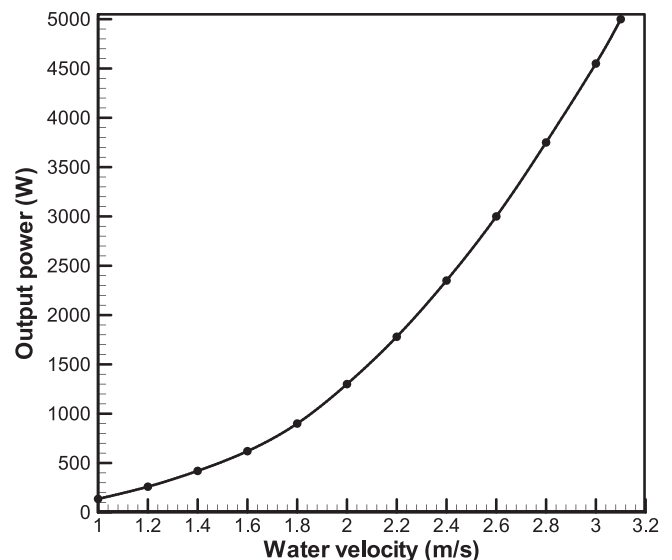


Fig. 7. Power curve of the considered hydrokinetic turbine.

Table 3

The power system techno-economic data.

<i>PV unit</i>	
Capital cost[\$/kW]	2000
Operation and maintenance[\$/year]	10
Lifetime [years]	25
Derating factor [%]	90
Ground reflection [%]	20
<i>Wind turbine system</i>	
Rated power [kW]	10
Capital cost [\$/unit]	20,000
Operation and maintenance cost [\$ /year]	400
Lifetime [year]	25
Hub height [m]	30
<i>Diesel generator</i>	
Capital cost [\$ /kW]	220
Replacement cost [\$ /kW]	200
Lifetime [hours]	15,000
Operation and maintenance [\$ /hour]	0.03
Diesel fuel price [\$ /L]	0.50
<i>Batteries (Generic 1 kWh Lead Acid)</i>	
Nominal voltage [V]	12
Nominal capacity [kWh]	1
Maximum capacity [Ah]	83.4
Maximum charge current [A]	16.7
Minimum state of charge [%]	40
Cost [\$ /unit]	110
Operation and maintenance cost [\$ /year]	10
Replacement cost [\$ /unit]	100
Lifetime [year]	10
<i>Converter</i>	
Capital cost [\$ /kW]	700
Replacement cost [\$ /kW]	630
Operation and maintenance [\$ /year]	10
Lifetime [years]	15
Efficiency [%]	95
<i>Hydrokinetic turbine</i>	
Manufactory	Smart Hydro Power
Rotational speed (rpm)	90 – 230
Weight (kg)	300
Number of rotor blades	3
Rotor diameter (mm)	1000
Capital cost[\$/unit]	13,500
Operation and maintenance[\$/year]	100
Replacement[\$/year]	400
Max. Power output	3.1 m/s – 5 kW

battery. The hydrokinetic turbine is a comparatively fresh innovation, so, The two options in this study will be based on it.

3.2.1. First option (PV-WT-DG-battery)

The hybrid power system for the RO desalination unit proposed in the first option does not include a hydrokinetic turbine. The hybrid power system that used to operate the RO desalination unit consists of various components; DG, WT, PV panels, converter, and battery banks.

Table 4

Different cases for the two options analyzed in HOMER.

Options	Cases	Hybrid system description	Number of Components and ratings (kW)
First Option	Case (1)	DG	DG (4.90 kW)/1 Battery/Converter (0.0365 kW)
	Case (2)	WT	9 wt (10 kW each)/ 940 batteries/ Converter (17.1 kW)
	Case (3)	PV	PV (42.2 kW)/421 batteries/Converter (7.49 kW)
	Case (4)	PV + WT	PV (26.4 kW)/2 wt (10 kW each)/290 batteries/Converter (5.04 kW)
	Case (5)	PV + DG	PV (0.919 kW)/ DG (4.90 kW)/1 battery/Converter (0.476 kW)
	Case (6)	WT + DG	1 wt (10 kW)/ DG (4.90 kW)/15 batteries/ Converter (1.93 kW)
	Case (7)	PV + WT + DG	PV (8.3 kW)/1 wt (10 kW)/ DG (4.90 kW)/15 batteries /Converter (5.29 kW)
Second Option	Case (1)	DG	DG (4.90 kW)/1 Battery/Converter (0.0365 kW)
	Case (3)	PV	PV (42.2 kW)/421 batteries/Converter (7.49 kW)
	Case (5)	PV + DG	PV (0.919 kW)/ DG (4.90 kW)/1 battery/Converter (0.476 kW)
	Case (8)	HKT	4 HKT (5 kW each)/ 20 batteries /Converter (0.00547 kW)
	Case (9)	PV + HKT	PV (0.138 kW)/4 HKT (5 kW each)/15 batteries/Converter (0.00273 kW)
	Case (10)	HKT + DG	3 HKT (5 kW each)/ DG (4.90 kW)/15 batteries/ Converter (2.24 kW)
	Case (11)	PV + HKT + DG	PV (2.82 kW)/ 3 HKT (5 kW each)/ DG (4.90 kW)/15 batteries /Converter (0.984 kW)

This energy system covers the energy required to operate the proposed RO desalination unit, and the excess power from the hybrid power system may be used to charge the battery bank. This option is the same cases that studied by Gökçek [2].

3.2.2. Second option (PV-HKT-DG- Battery)

From the current study, it is reasonable that the wind turbine component has the highest capital and total cost. Therefore, it is proposed to replace the wind turbine component by a hydrokinetic turbine and investigate the economic analysis of the proposed model.

4. Results and discussion

4.1. Results and discussion of the first option

In water treatment processes to become drinkable like reverse osmosis systems, the main factor of these plants is energy consumption. The RO desalination system in which the energy consumption of freshwater ability of 24 m³/day concentrated in the current work is computed to become 4.38 kWh/m³. The selection of the capacities for the parts utilized within a power system highly influences the overall energy output, capital cost, and cost of the electricity. The first option hybrid energy system components are; WT, PV panel, DG, and battery banks for RO desalination unit might have been displayed in HOMER software utilizing various simulations. HOMER software modeled all of the potential integration as stated by the input parameters and required optimization results from the most reduced NPC (Net Present Cost) with the most elevated NPC results. This option investigated the integration of seven various energy configurations for supplying power to the RO desalination unit as shown in Table 4. Modeling and simulations were carried out where the diesel generator system alone (Case 1), wind turbine system alone (Case 2), PV solar system alone (Case 3), the hybrid system composed of PV solar panels wind turbine (Case 4), a hybrid system composed of PV solar panels with diesel generator (Case 5), a hybrid system composed of wind turbine with diesel generator (Case 6), and a hybrid system composed of PV solar panels, a wind turbine with diesel generator (Case 7). All hybrid power systems considered in the modeling and simulation processes contain batteries that act as a power storage element taking into account the project lifetime of 25 years.

Fig. 8 indicates the net present cost of all proposed cases of the first option from Case 1 to Case 7. Case 1 is used of a diesel generator system only, which is a popular system used for off-grid electricity generation for RO desalination unit. It was found that the diesel generator (Case 1) is the lowest capital expensive compared to other elements where the cost is (1214\$) from the results of modeling and simulation. But in the case of a diesel generator alone, the cost of fuel must be taken into account as the price of fuel has a considerable impact on the economy

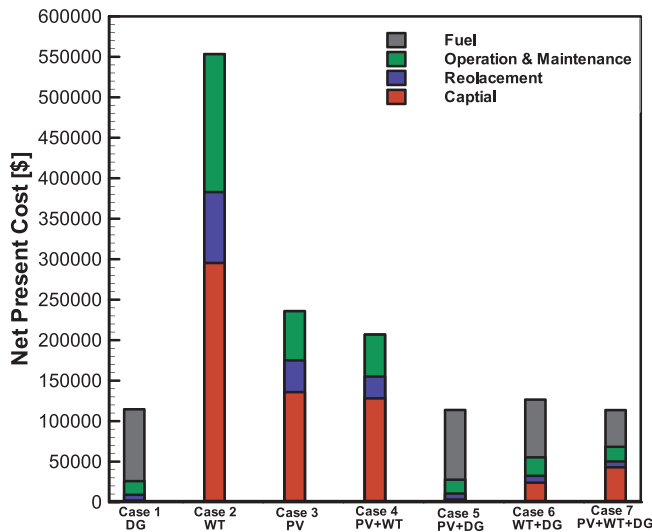


Fig. 8. Cash flow summary of various components for the best optimized hybrid system for the first option.

side. In the current study, the price of fuel utilized in diesel generator was considered to be 0.55 \$ /L in Egypt. From Table 4 it was found that Case 2 is the use of a wind turbine system only as a power source. The wind turbine system has the capital cost is 295,363 \$, which is high relative to the cost of other systems. Other cases from Case 3 to Case 7 have capital costs more than the minimum value of Case 1 and lower than the maximum value of Case 2.

Fig. 9 indicates the levelized cost of energy (LCOE) for all cases considered in the first option. This figure shows that Case 7 has the minimum value of LCOE is 0.2252 \$/kWh. On the other hand, Case 2 has the maximum value of LCOE is 1.09 \$/kWh due to the wind turbine has the highest capital cost. Also, Case 5 has LCOE value of 0.2292 \$/kWh, which is the nearest one to Case 7.

Table 5 indicates the optimized results from the simulation cases in the first option. In Case 1 where only the diesel generator is used to generate the power needed to operate the RO desalination unit, the power of the diesel generator is 4.90 kW, producing 38,369 kWh. It also has a normal electrical efficiency of 34% with no excess of generated electricity. For (Case 2) WT system was used only, the power generation per year would be 106,333 kWh/year. In Case 2, the optimal system consists of 9 wt (10 kW each), 17.1 kW converter, and 940 kWh LA batteries with 58% excess of generated power. Furthermore, for (Case 3) PV solar panels system was used only, the power generation per year would be 79,258 kWh/year. In Case 3, the optimal system consists of 42.2 kW PV panels, 7.49 kW converter, and 421 kWh LA batteries with 42.3% excess of generated power. For (Case 4) hybrid system PV-WT was used, the power generation per year would be 73,190 kWh/year, whereas PV panels produced 67.72% (49,561 kWh/year) of the total power and WT produced 32.28% (23,629 kWh/year) of the total power. Based on the results of modeling and simulation using the HOMER software, the optimal hybrid system PV-WT (Case 4) consists of 26.4 kW PV panels, 2 wt (10 kW each), 5.04 kW converter, and 290 kWh LA batteries with 40.2% excess of generated power. For (Case 5) hybrid system PV-DG was used, the power generation per year would be 38,694 kWh/year, whereas PV panels produced 4.5% (1728 kWh/year) of the total power and DG produced 95.5% (36,966 kWh/year) of the total power. The optimal hybrid system PV-DG (Case 5) consists of 0.919 kW PV panels, 4.9 kW DG, 0.476 kW converter, and 1 kWh LA batteries with 7.13% excess of generated power. For (Case 6) hybrid system WT-DG was used, the power generation per year would be 41,315 kWh/year, whereas WT produced 28.6% (11,815 kWh/year) of the total power and DG produced 71.4% (29,500 kWh/year) of the total power. The optimal hybrid system WT-DG (Case 6) consists of one WT

(10 kW), 4.9 kW DG, 1.93 kW converter, and 15 kWh LA batteries with 0.649% excess of generated power. In Cases 5 and 6, most of the power is generated by the diesel generator, with renewable portions for 3.7% and 23.1%, respectively. For (Case 7) hybrid system PV-WT-DG was used, the power generation per year would be 47,207 kWh/year. The optimal hybrid system PV-WT-DG (Case 7) consists of 8.3 kW PV panels, one WT (10 kW), 4.9 kW DG, 5.29 kW converter, and 15 kWh LA batteries with 17.4% excess of generated power. The best choice of the recognized cases in the first option is Case 7. In this case, the levelized cost of electricity is the smallest value of 0.2252 \$/kWh as shown in Fig. 9.

Air pollutants are emitted from the fuel burning process inside the diesel generator, which is part of the hybrid power system. The most important air pollutants emitted from the diesel generator are CO, CO₂, NO_x, SO₂, particulate matter and unburned hydrocarbons. Consequently, it is important to evaluate the operational emanations of the hybrid power systems which include diesel generator which is the source of pollutant emissions. Table 6 indicates the pollutant emissions from different cases considered in this system. The worst choice from the various system considered is the diesel system alone (Case 1) where the power generation only depends on the fossil fuel burned, so, it has maximum pollutant emissions (30,329.8 kg/year). While the suggested optimal hybrid power system (Case 7) contains air pollutant emissions, but it has the smallest value of pollutants between systems utilizing diesel generator (15,907.62 kg/year). Also, the pollution emissions from the power system depend on the hours of working for the diesel generator. As the number of hours of working for the diesel increase, the pollution emissions released to the environment increases also.

Fig. 10 indicates the power produced in each month by every component of the optimized hybrid system in Case 7. For the renewable energy system, power production has discontinuous performance throughout the year due to the nature of renewable sources. As stated by the results from Case 7, the wind turbine work depends on the wind speed which has a small change from one month to another through the year. So, The wind turbine has a maximum power generation in February and March with maximum wind speed. But the wind turbine has minimum power generation in August which has the minimum wind speed. The wind turbine power generated in a year (11,468 kWh/year) which equal to 24.3% forms the annual power generated from the hybrid system. The PV panels work relies on solar radiation and clearness index. Therefore, the electricity generated by the PV panels is maximum in June and July with a maximum solar radiation and clearness

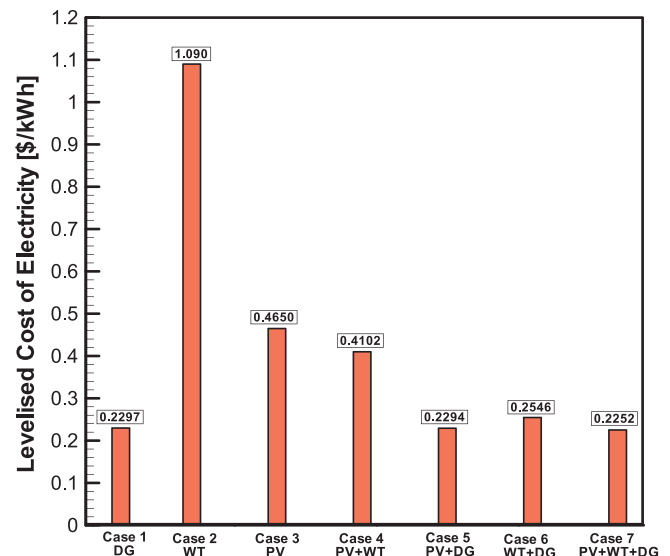


Fig. 9. Cost of electricity for the best optimized hybrid system for the first option..

Table 5
Optimized cases from the first option.

	Case 1 DG	Case 2 WT	Case 3 PV	Case 4 PV + WT	Case 5 PV + DG	Case 6 WT + DG	Case 7 PV + WT + DG
PV (kW)	–	–	42.2	26.4	0.919	–	8.3
WT (number)	–	9	–	2	–	1	1
Diesel Generator (kW)	4.9	–	–	–	4.9	4.9	4.9
Battery (1 kWh LA)	1	940	421	290	1	15	15
Converter (kW)	0.0365	17.1	7.49	5.04	0.476	1.93	5.29
PV electricity (kW/year)	–	–	79,258	49,561	1,728	–	15,602
WT electricity (kW/year)	–	106,333	–	23,629	–	11,815	11,468
Diesel Generator electricity (kW/year)	38,369	–	–	–	36,966	29,500	20,137
Excess electricity (%)	0	58	42.3	40.2	0.649	7.13	17.4
Renewable Fraction (%)	0	100	100	100	3.7	23.1	52.5

Table 6
Pollutants emission from the first option cases.

Pollutants	Case 1 (kg/year)	Case 5 (kg/year)	Case 6 (kg/year)	Case 7 (kg/year)
Carbon dioxide	29,980	29,114	24,064	15,658
Carbon monoxide	189	184	152	105
Unburned hydrocarbons	8.25	8.01	6.62	4.58
Particulate matter	1.15	1.11	0.919	0.636
Sulfur dioxide	73.4	71.3	58.9	40.8
Nitrogen oxides	178	172	142	98.6
Total emission	30,329.8	29,550.42	24,424.44	15,907.62

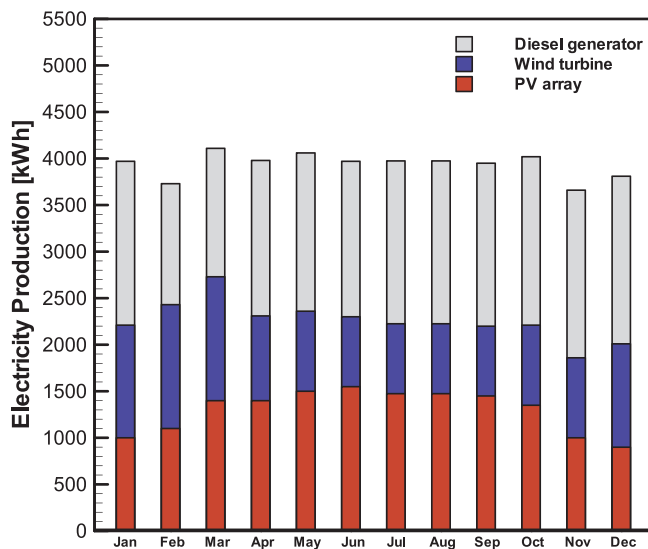


Fig. 10. Monthly average electricity production for system considered in Case (7).

index. Also, December has minimum solar radiation and clearness index. As a result, power produced by PV solar panels is the minimum value through this month. The PV solar panels power generated in a year (15,602 kWh/year) which is equal to 33.1% forms the annual electricity generated from the hybrid system. Furthermore, the integration of the diesel generator with renewable components is required to complete the total demand load of electricity from the hybrid power system. The lowest value of the diesel generator power generation was in February and March. The power generated by the diesel generator in a year (20,137 kWh/year) which equal to 42.6% form the annual electricity generated from the hybrid system.

Fig. 11 indicates the quantity of fuel that is burned inside the diesel generator for the optimum power production system (Case 7). The minimum fuel burned in a diesel generator in February (410.8 L). The annual fuel used in the diesel generator is 6364 L/year as shown in

Figs. 12 and 13. As renewable power production accessibility decreases, the fuel consuming increases.

The leveled cost method is the major essential strategy for evaluation the energy cost as well as the cost of water produced. It is popular utilized to produce standards to define the feasibility of various power production technologies [38]. Table 7 indicates the Levelized cost of water for RO desalination unit for hybrid power generation systems in all forms available in the first option for the current work and Gökçek [2]. Table 7 indicates that for the current study, the smallest value of LCOW is 1.10 \$/m³ for Case 7 which consists of PV, WT, DG, and backup battery hybrid power system. For Gökçek [2], the smallest value of LCOW is 2.20 \$/m³ for Case 7 which consists of PV, WT, DG, and backup battery hybrid power system. The highest value of LCOW is 7.67 \$/m³ for the current study but the highest value of LCOW is 7.67 \$/m³ for Gökçek [2] that occurs in Case 2, which consists of WT system only. It is clear from Table 7 that cases contains diesel generator; Case (1), Case (5), Case (6), and Case (7) the LCOW for the present study is less than LCOW for Gökçek [2] due to the difference in fuel price. But other cases that not contain diesel generator; Case (2), Case (3), and Case (4) the LCOW for the present study is different from Gökçek [2] due to the difference in wind speed, solar irradiance, and clearness index.

4.2. Results and discussion of second option

The second option hybrid energy system components are; HKT, PV solar cells, diesel generators, and battery might have been displayed in

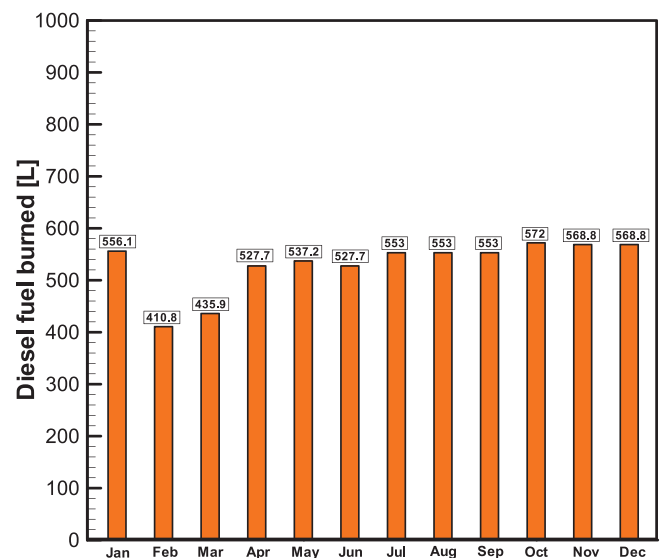


Fig. 11. The monthly amount of fuel consumed by the diesel generator for system considered in Case (7).

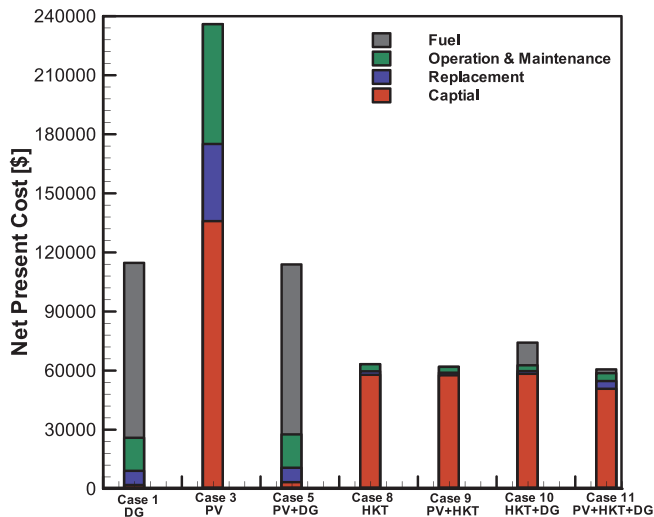


Fig. 12. Cash flow summary of various components for the best optimized hybrid system for the second option.

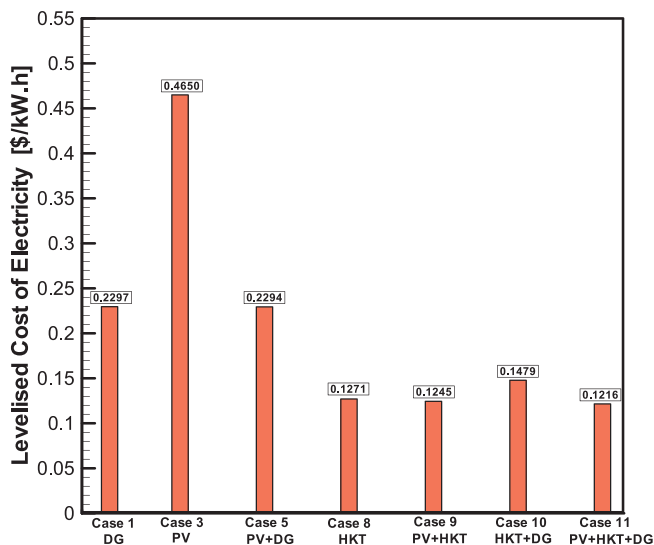


Fig. 13. Cost of electricity for the best optimized hybrid system for the second option.

HOMER software utilizing various simulations. The second option investigated the integration of seven various power configurations to supply power for RO desalination unit as shown in Table 4.

Modeling and simulations were carried out where the diesel generator system alone (Case 1), PV solar system alone (Case 3), hybrid system composed of PV solar panels and diesel generator (Case 5), hydrokinetic turbine system alone (Case 8), the hybrid system composed of PV solar panels and hydrokinetic turbine (Case 9), hybrid system composed of hydrokinetic turbine and diesel generator (Case 10), and hybrid system composed of PV solar panels, hydrokinetic turbine and diesel generator (Case 11). Cases 1, 3, and 5 are the same case from the previous investigations in the first option. The power generation systems considered in Cases 8, 9, 10, and 11 will be

investigated due to HKT installed in the hybrid system.

Fig. 12 indicates the NPC results of the optimized simulations for cases considered in the second option. The HKT system considered in Case 8 was utilized for off-grid power generation for RO desalination unit. As stated by simulation results, this case has capital cost (57,884 \$) which is cheaper than the wind turbine. The HKT-PV hybrid system is considered in Case 9 and the HKT-DG is considered in Case 10. The HKT-PV-DG is simulated in Case 11. As clearly shown in Fig. 12, Case 3 which has PV panels has the maximum capital expense compared to other arrangements (135,899 \$). Fig. 13 indicates the LCOE values for all cases considered in the second option. It is clear that the minimum LCOE value in Case 11 of 0.1216 \$/kWh, while the maximum LCOE value in Case 3 of 0.465 \$/kWh. Also, the HKT-PV hybrid system (Case 9) has LCOE value of 0.1245 \$/kWh, which is the nearest one to Case 11.

Table 8 presented the results of optimized cases gotten from the simulation of the second option. For (Case 8) HKT system was used only, the power generation per year would be 45,552 kWh/year. In Case 8, the optimal system consists of 4 HKT, 0.00547 kW converter, and 20 kWh LA batteries with 15.8% excess of generated power. For (Case 9) hybrid system PV-HKT was used, the power generation per year would be 45,811 kWh/year, whereas PV panels produced 0.56% (259 kWh/year) of the total power and HKT produced 99.44% (45,552 kWh/year) of the total power. Based on the results of modeling and simulation using the HOMER software, the optimal hybrid system PV-HKT (Case 9) consists of 0.138 kW PV panels, 4 HKT, 0.00273 kW converter, and 15 kWh LA batteries with 16.2% excess of generated power. For (Case 10) hybrid system HKT-DG was used, the power generation per year would be 39,607 kWh/year, whereas HKT produced 86.26% (34,164 kWh/year) of the total power and DG produced 13.74% (5,443 kWh/year) of the total power. The optimal hybrid system HKT-DG (Case 10) consists of one 3 HKT, 4.9 kW DG, 2.24 kW converter, and 15 kWh LA batteries with no excess of generated power. For (Case 11) hybrid system PV-HKT-DG was used, the power generation per year would be 39,982 kWh/year. The optimal hybrid system PV-HKT-DG (Case 11) consists of 2.82 kW PV panels, 3 HKT, 4.9 kW DG, 0.984 kW converter, and 15 kWh LA batteries with 2.02% excess of generated power. The best choice of the recognized cases in the second option in Case 11. In this case, the levelized cost of electricity is the smallest value of 0.1216 \$/kWh as shown in Fig. 13.

Table 9 indicates the pollutant emissions from different cases considered in the second option. The suggested hybrid configuration (Case 11) has the minimum pollutant emissions when compared with all cases (contained diesel generator) considered in the current study.

Fig. 14 indicates the power produced in each month by every component of the optimized hybrid system in Case 11. According to the results for Case 11, the electricity from the PV panels is small when compared with HKT throughout all the months of the year. Furthermore, the diesel generator is a very small electricity production throughout the year. Throughout June, July, and August there is no need to operate the diesel generator. The electricity generated by HKT in a year (34,164 kWh/year) which is equal to 85.45% of the total annual generation power. The power produced by the PV panels in a year (5,291 kWh/year) which is equal to 13.23% of the total annual generation power. The power produced by the diesel generator in a year (527 kWh/year) which is equal to 1.32% forms the annual electricity generated from the hybrid system.

Fig. 15 indicates the quantity of fuel that is burned inside the diesel

Table 7

Levelised cost of water for cases in the first option.

		Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
LCOW(\$/m ³)	Present study	1.123	7.67	4.21	3.42	1.121	1.23	1.10
	Gökçek [2]	3.22	5.11	3.75	2.27	2.41	2.69	2.20

Table 8
Optimized cases from the second option.

	Case 1 DG	Case 3 PV	Case 5 PV-DG	Case 8 HKT	Case 9 PV-HKT	Case 10 HKT-DG	Case 11 PV-HKT-DG
PV (kW)	–	42.2	0.919	–	0.138	–	2.82
HKT (number)	–	–	–	4	4	3	3
Diesel Generator (kW)	4.9	–	4.9	–	–	4.9	4.9
Battery (1 kWh LA)	1	421	1	20	15	15	15
Converter (kW)	0.0365	7.49	0.476	0.00547	0.00273	2.24	0.984
PV electricity (kW/year)	–	79,258	1,728	–	259	–	5291
HKT electricity (kW/year)	–	–	–	45,552	45,552	34,164	34,164
Diesel Generator electricity (kW/year)	38,369	–	36,966	–	–	5443	527
Excess electricity (%)	0	42.3	0.649	15.8	16.2	0	2.02
Renewable Fraction (%)	0	100	3.7	100	100	85.8	98.2

Table 9
Pollutants emission from the second option cases.

Pollutants	Case 1 DG (kg/year)	Case 5 PV + DG (kg/year)	Case 10 HKT + DG (kg/year)	Case 11 PV + HKT + DG (kg/year)
Carbon dioxide	29,980	29,114	4810	634
Carbon monoxide	189	184	30.3	3.99
Unburned hydrocarbons	8.25	8.01	1.32	0.174
Particulate matter	1.15	1.11	0.184	0.0242
Sulfur dioxide	73.4	71.3	11.8	1.55
Nitrogen oxides	178	172	28.5	3.75
Total emission	30,429.8	29,550.42	4,882.1	643.49

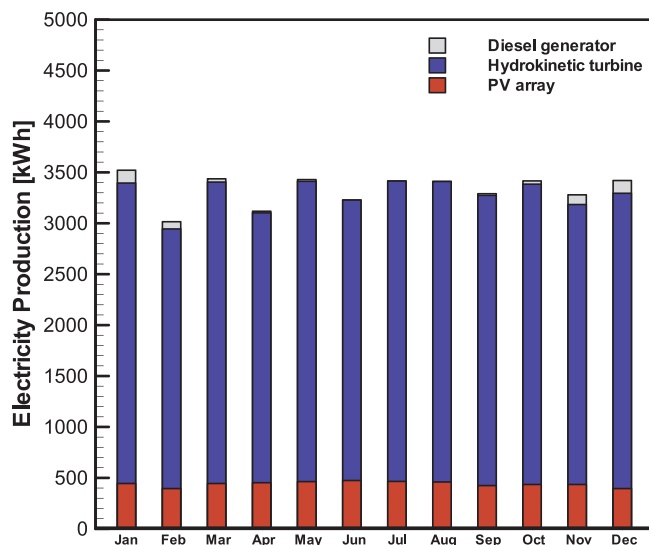


Fig. 14. Monthly average electricity production for system considered in Case (11).

generator for the optimum power production system (Case 11). There is no fuel burned through the diesel generator in June, July, and August where there is no load on the diesel generator. The fuel utilized in the diesel generator is 242 L/year.

LCOW for the RO desalination unit operated by the hybrid systems with the seven cases proposed in the second option are listed in Table 10. The smallest value of LCOW is 0.56 \$/m³ for Case 11 which consists of PV, HKT, DG, and backup battery hybrid power system. However, the highest value of LCOW is 4.21 \$/m³ for Case 3, which consists of PV system only.

To investigate the effect of the diesel fuel price on the desalinated water production cost for the present study is carried out. Fig. 16 shows the LCOW values for the optimum cases of the present study and compared with Gökçek [2] with the change of fuel price. From Fig. 16 it

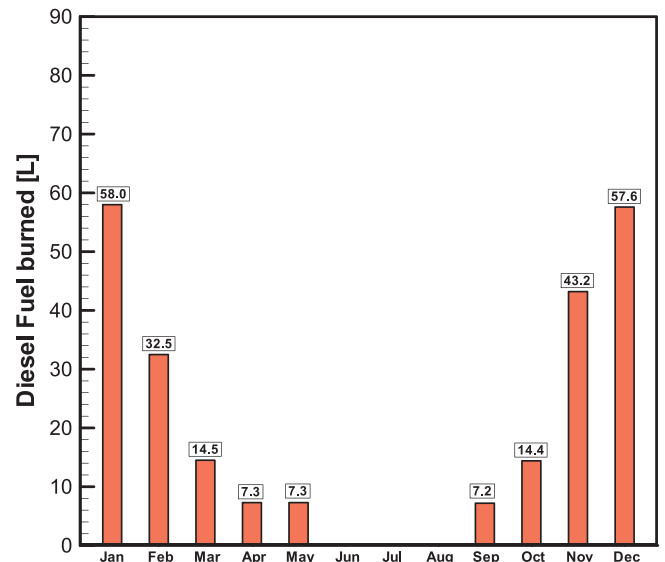


Fig. 15. The monthly amount of fuel consumed by the diesel generator for system considered in Case (11)

is clear that the increase in fuel price causes an increase in the LCOW value. The optimum case containing the hydrokinetic turbine is the lowest LCOW when compared with the other cases.

5. Conclusions

The energy required for water treatment systems in Egypt can be obtained from hybrid energy systems based on renewable energy sources. The current study estimated several energy production systems with elements; photovoltaic solar panel, wind turbine, hydrokinetic turbine, and diesel generator with a backup battery system for RO desalination unit produces 24 m³/day for utilized in Ras El Bar, Egypt. The power generation system in the current study includes two options; the first option components; photovoltaic solar panel, wind turbine, and diesel generator while the second option components photovoltaic solar panel, hydrokinetic turbine, and diesel generator. Some important points can be summarized from this work as follow:

- The current study proposed two options of a hybrid energy system to

Table 10
Levelised cost of water for cases in the second option.

	Case 1 DG	Case 8 HKT	Case 3 PV	Case 9 PV-HKT	Case 5 PV-DG	Case 10 HKT-DG	Case 11 PV-HKT-DG
LCOW(\$/m ³)	1.123	0.66	4.21	0.65	1.121	0.77	0.56

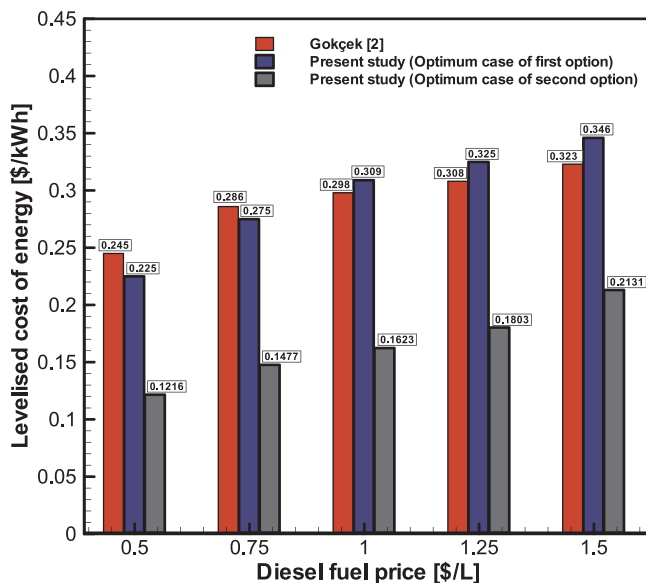


Fig. 16. LCOW values for different fuel price.

provide energy for RO desalination unit. The first option components; photovoltaic solar panel, wind turbine, and diesel generator with optimized hybrid system which included 8.3 kW PV, 1 wt (10 kW rated power), 4.90 kW DG, 5.29 kW converter, and 15 kWh LA battery (Case 7). While the second option components; photovoltaic solar panel, hydrokinetic turbine, and diesel generator with optimized hybrid system which included 2.8 kW PV, 3 HKT (5 kW maximum power), 4.90 kW DG, 0.984 kW converter, and 15 kWh LA battery (Case 11).

- The energy cost of the optimal hybrid system of the first option (Case 7) was 0.2252 \$/kWh. In the second option, the energy cost of the optimal hybrid system (Case 11) was 0.1216 \$/kWh.
- In the first option, the value of net present cost and renewable energy fraction for the optimal case (Case 7) is 119,260 \$ and 52.5%, respectively. In the second option, the value of net present cost and renewable energy fraction for the optimal case (Case 11) are 60,333 \$ and 98.6%, respectively.
- The LCOW was 1.10 \$/m³ for the first option (Case 7) while The LCOW was 0.56 \$/m³ for the second option (Case 11).
- The worst choice of the hybrid power generation system is the DG alone (Case 1) which has maximum emissions with CO₂ emissions are 3029.8 kg/year.
- Pollutant emissions in the optimal case (Case 11) with CO₂ emissions are 643.48 kg/year as annual CO₂ emissions are reduced by 97.8%.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Khan MA, Rehman S, Al-Sulaiman FA. A hybrid renewable energy system as a potential energy source for water desalination using reverse osmosis: A review. *Renew Sustain Energy Rev* 2018;97:456–77.
- [2] Gökçek M. Integration of hybrid power (wind-photovoltaic-diesel-battery) and seawater reverse osmosis systems for small-scale desalination applications. *Desalination* 2018;435:210–20.
- [3] Gilau AM, Small MJ. Designing cost-effective seawater reverse osmosis system under optimal energy options. *Renewable Energy* 2008;33:617–30.
- [4] Mousa K, Diabat A, Fath H. Optimal design of a hybrid solar-wind power to drive a small-size reverse osmosis desalination plant. *Desalination* 2013;31:3417–27.
- [5] Nagaraj R, Thirugnanamurthy D, Rajput MM, Panigrahi BK. Techno-economic

- analysis of hybrid power system sizing applied to small desalination plants for sustainable operation. *Int J Sustain Built Environ* 2016;5:269–76.
- [6] Salehin S, Ferdaous MT, Chowdhury RM, Shithi SS, Rofi MB, Mohammed MA. Assessment of renewable energy systems combining techno-economic optimization with energy scenario analysis. *Energy* 2016;112:729–41.
- [7] Koroneos C, Dompros A, Roumbas G. Renewable energy driven desalination systems modelling. *J Cleaner Prod* 2007;15:449–64.
- [8] Setiawan AA, Zhao Y, Nayar CV. Design, economic analysis and environmental considerations of mini-grid hybrid power system with reverse osmosis desalination plant for remote areas. *Renewable Energy* 2009;34:374–83.
- [9] Spyrou ID, Anagnostopoulos JS. Design study of a stand-alone desalination system powered by renewable energy sources and a pumped storage unit. *Desalination* 2010;257:137–49.
- [10] Cherif H, Belhadi J. Large-scale time evaluation for energy estimation of standalone hybrid photovoltaic-wind system feeding a reverse osmosis desalination unit. *Energy* 2011;36:6058–67.
- [11] Khalifa AN. Evaluation of different hybrid power scenarios to Reverse Osmosis (RO) desalination units in isolated areas in Iraq. *Energy Sustain Dev* 2011;15:49–54.
- [12] Hossam-Eldin A, El-Nashar AM, Ismaiel A. Investigation into economical desalination using optimized hybrid renewable energy system. *Electr Power Energy Syst* 2012;43:1393–400.
- [13] Mentis D, Karalis G, Zervos A, Howells M, Taliotis C, Bazilian M, et al. Desalination using renewable energy sources on the arid islands of South Aegean Sea. *Energy* 2016;94:262–72.
- [14] Javed MS, Song A, Ma T. Techno-economic assessment of a stand-alone hybrid solar-wind-battery system for a remote island using genetic algorithm. *Energy* 2019;176:704–17.
- [15] Krishan O, Suhag S. Techno-economic analysis of a hybrid renewable energy system for an energy poor rural community. *J Storage Mater* 2019;23:305–19.
- [16] Kusakana K. Techno-economic analysis of off-grid hydrokinetic-based hybrid energy systems for onshore/remote area in South Africa. *Energy* 2014;68:947–57.
- [17] Kusakana K, Vermaak HJ. Cost and performance evaluation of hydrokinetic-diesel hybrid systems. *Energy Procedia* 2014;61:2439–42.
- [18] Lata-García J, Jurado F, Fernandez-Ramírez LM, Sanchez-Sainz H. Optimal hydrokinetic turbine location and techno-economic analysis of a hybrid system based on photovoltaic/hydrokinetic/hydrogen/battery. *Energy* 2018;159:611–20.
- [19] Aziz AS, Tajuddin MN, Adzman MR, Azmi A, Ramli MM. Optimization and sensitivity analysis of standalone hybrid energy systems for rural electrification: A case study of Iraq. *Renewable Energy* 2019;138:775–92.
- [20] Gökçek M, Gökçek ÖB. Technical and economic evaluation of freshwater production from a wind-powered small-scale seawater reverse osmosis system (WP-SWRO). *Desalination* 2016;381:47–57.
- [21] HOMER Pro Version 3.7 User Manual, August 2016, www.homerenergy.com.
- [22] NASA, 2015. Surface Meteorology and Solar Energy available at: <https://eosweb.larc.nasa.gov/cgi-bin/sse/grid.cgi?email=skip@larc.nasa.gov> (accessed February 2015).
- [23] Eshra NM, Abdelnaby ME. Study of Application of Small Hydropower for Nile River in Egypt. *Mid East J Appl Sci* 2014;4(4):1118–29.
- [24] Torunoğlu PO, Orhon D. Cost analyses based on scientific design for salinity water membrane technology systems can be installed in Turkey. *Itüjournal/e Water Pollut Control* 2010:97–110.
- [25] Mohammed A, Pasupuleti J, Khatib T, Elmenreich W. A review of process and operational system control of hybrid photovoltaic/diesel generator systems. *Renew Sustain Energy Rev* 2015;44:436–46.
- [26] Duffie JA, Beckman WA. *Solar Engineering of Thermal Processes*. fourth ed. Wiley; 2013.
- [27] Baneshi M, Hadianfard F. Techno-economic feasibility of hybrid diesel/PV/wind/battery electricity generation systems for non-residential large electricity consumers under southern Iran climate conditions. *Energy Convers Manage* 2016;127:233–44.
- [28] Kumar D, Sarkar S. A review on the technology, performance, design optimization, reliability, techno-economics and environmental impacts of hydrokinetic energy conversion systems. *Renew Sustain Energy Rev* 2016;58:796–813.
- [29] Bernal-Agustín JL, Dufo-Lopez R. Simulation and optimization of stand-alone hybrid renewable energy systems. *Renew Sustain Energy Rev* 2009;13:2111–8.
- [30] Yuce MI, Muratoglu A. Hydrokinetic energy conversion systems: A technology status review. *Renew Sustain Energy Rev* 2015;43:72–82.
- [31] Zahboun H, Zouggar S, Krajacic G, Varbanov PS, Elhafyani M, Ziani E. Optimal hybrid renewable energy design in autonomous system using Modified Electric System Cascade Analysis and Homer software. *Energy Convers Manage* 2016;126:909–22.
- [32] Amutha WM, Rajini V. Cost benefit and technical analysis of rural electrification alternatives in southern India using HOMER. *Renew Sustain Energy Rev* 2016;62:236–46.
- [33] Hossain M, Mekhilef S, Olatomiwa L. Performance evaluation of a stand-alone PV-wind-diesel-battery hybrid system feasible for a large resort center in South China Sea. *Malaysia. Sustainable Cities and Society* 2017;28:358–66.
- [34] Isa NM, Das HS, Tan CW, Yatim AM, Lau KY. A techno-economic assessment of a combined heat and power photovoltaic/fuel cell/battery energy system in Malaysia hospital. *Energy* 2016;112:75–90.
- [35] Brandoni C, Bosnjakovic B. HOMER analysis of the water and renewable energy nexus for water-stressed urban areas in sub-Saharan Africa. *J Cleaner Prod* 2017;155:105–18.
- [36] Halabi LM, Mekhilef S, Olatomiwa L, Hazelton J. Performance analysis of hybrid PV/diesel/battery system using HOMER: a case study Sabah. *Malaysia. Energy Conversion and Management* 2017;144:322–39.
- [37] Tazvinga H, Xia X, Zhang J. Minimum cost solution of photovoltaic diesel battery hybrid power systems for remote consumers. *Sol Energy* 2013;96:292–9.
- [38] Vazquez A, Iglesias G. LCOE (levelised cost of energy) mapping: a new geospatial tool for tidal stream energy. *Energy* 2015;91:192–201.